



Cambrian College

School of Business and Information Technology, Creative Arts, Design, Music and Hospitality

Course Outline

Course Title	Software Engineering Concepts				
Course Code:	STY1001	Credit Value:	4	Credit Hours:	56
Programs:	BAGC I.T. Business Analysis				
Equivalencies		Prerequisites		Corequisites	

This course may be delivered in a variety of different formats: 100% in-class, 100% online (or a blend of both), videoconferencing, distributed learning or off-campus. Please confirm with your faculty member which format will be used for your section of this course.

General Education Course: ☐

Eligible for PLAR: ☒

COURSE DESCRIPTION

In this course, students will investigate and apply various approaches to problem-solving systems. The student will solve simple problems in code and will then consider how to break more extensive problems down into a series of smaller components. Students will evaluate and create simple applications and work with various data structures. Finally, the student will work with custom, programmer-defined objects and different types of connectivity.

Date: July 09, 2020

Prepared By: Nathan Abourbih

Approved by: 

BRIAN LOBBAN
Dean, School Of Business And Information Technology,
Creative Arts, Design, Music And Hospitality

Effective: Fall 2020, Winter 2021, Spring 2021

RELATIONSHIP TO PROGRAM VOCATIONAL LEARNING OUTCOMES

PROGRAM LEVEL	
This course contributes to your program by allowing you to demonstrate the following vocational learning outcomes:	
Program(s)	Vocational Learning Outcomes
I.T. Business Analysis	College Standards 1. Analyze the information collected during the requirements elicitation to identify gaps and define the capabilities of an acceptable solution using appropriate methods, tools, techniques and documentation. 2. Document business requirements in accordance with standard business practices.

COURSE CURRICULUM**Topics/Concepts Covered in This Course**

- Software Engineering
- Software Programming
- Integrated Development Environments
- Conventions, Whitespaces, and Comments
- Literal and Variable Values
- Class Libraries
- Decision Making
- Sequences of Data
- Connectivity
- Debugging and Testing
- Online Resources and Collaboration

COURSE LEVEL: Learning Outcomes and Objectives	
To earn credit for this course, you must reliably demonstrate your ability to:	
Learning Outcome	Objectives
1. Discuss the core concepts of software engineering.	 1.1 Explain the importance of gathering, analyzing, specifying, and validating the requirements of the software. 1.2 Describe software development by summarizing the steps of designing, programming, and documenting. 1.3 Recognize the importance of evaluating the functionality of software to determine if it meets the requirements or to determine if it contains any defects. 1.4 Discuss the process of improving and fixing software problems after it has already been delivered.

Learning Outcome	Objectives
2. Describe the fundamentals of modern software programming.	2.1 List and describe the most commonly used programming paradigms. 2.2 Describe what a programming language is and the importance of syntax. 2.3 Explain what code compilation is and why you need to compile some types of code and not others. 2.4 Describe the core concepts of object-oriented programming using the following terms: classes, objects, inheritance, interfaces, generics, and delegates.
3. Create an application using an integrated development environment.	3.1 Describe the purpose and benefits of using an IDE when developing software. 3.2 Explain the developer workflow and identify important project components and their purpose. 3.3 Demonstrate the functions and capabilities of the code editor and visual editor. 3.4 Create a new project and describe the general structure of a C# application. 3.5 Use basic syntax rules of C# and describe the flow of execution. 3.6 Demonstrate running and debugging an application.
4. Write readable and well-organized code.	4.1 Create descriptive names for variables and classes. 4.2 Demonstrate the use of comments to describe the purpose of code and higher-level requirements. 4.3 Write code that effectively uses whitespaces and newlines. 4.4 Organize code and logic using classes and methods.
5. Demonstrate the use of literal and variable values.	5.1 List and describe the different kinds of value and reference types. 5.2 Explain the difference between a literal and a variable. 5.3 Declare and initialize variables. 5.4 Retrieve and set values in variables. 5.5 Create string data containing tabs, newlines, and other special characters. 5.6 Combine string data using concatenation and interpolation. 5.7 Perform mathematical operations on numeric values. 5.8 Write code that will explicitly and implicitly convert between different types.
6. Use methods from the .NET Class Library.	6.1 Write code that calls stateless methods in the .NET Class Library. 6.2 Create a new instance of .NET Class Library classes to call methods that maintain state.

Learning Outcome	Objectives
	6.3 Use Intellisense to learn more about a method, its overloaded versions, its return value data type, and its input parameter data types 6.4 Use docs.microsoft.com to research what a method does, its overloaded versions, its return value type, its input parameters, and what each parameter represents.
7. Add decision making logic to branch a code's execution path.	7.1 Describe the most widely used branching statements and provide use cases for each. 7.2 Write code that evaluates conditions using the if-elseif-else statements. 7.3 Build boolean expressions to evaluate conditions. 7.4 Combine boolean expressions using logical operators. 7.5 Nest code blocks within other code blocks. 7.6 Demonstrate the use of error checking and exception handling techniques.
8. Work with sequences of data stored in arrays and collections.	8.1 Explain the difference between arrays and collections. 8.2 Describe three kinds of collections and how to use them. 8.3 Create, initialize, and work with values in an array. 8.4 Define and work with custom collections. 8.5 Use LINQ to access a collection. 8.6 Iterate through sequences of data using the foreach statement. 8.7 Sort and manipulate sequences of data.
9. Create a highly-connected application.	9.1 Describe the core concepts of cloud computing. 9.2 Create a free Azure for Students account. 9.3 Provision an Azure SQL database to store application data. 9.4 Write code that consumes a Web API/SQL database. 9.5 Perform file I/O operations on the local filesystem. 9.6 Download a file from a given HTTP URL and save it. 9.7 Demonstrate the serialization and deserialization of an object.
10. Develop robust applications through debugging and software testing.	10.1 Describe the fundamentals of debugging and software testing. 10.2 Discuss the most common methods of debugging and software testing on this platform. 10.3 Select and use the most appropriate software testing techniques for a given piece of software to check for defects. 10.4 List and describe five rules that can result in clean tests. 10.5 Demonstrate the use of debugging, testing, and monitoring tools, services, and techniques.

Learning Outcome	Objectives
11. Use online resources effectively and collaborate with other software developers.	11.1 Discuss the challenges of using the Internet as a programming resource. 11.2 Identify and evaluate information found online to help solve programming problems. 11.3 List and describe the most common open-source licenses and their limitations. 11.4 Explain how to use someone else's code ethically. 11.5 Use open-source and third-party libraries/frameworks to develop an application's design/functionality. 11.6 Explain the purpose of version control using the following terms: repository, fork, fetch, branch, commit, clone, push, issue, merge, and blame. 11.7 Demonstrate the use of version control.

Essential Employability Skills

Communication

- communicate clearly in written, spoken, and visual form that fulfills purpose/needs of audience.
- respond to written, spoken, or visual messages in a manner that ensures effective communication.

Numeracy

- execute mathematical operations accurately.

Critical Thinking and Problem Solving

- apply a systematic approach to solve problems.
- use a variety of thinking skills to anticipate and solve problems.

Information Management

- locate, select, organize, and document information using appropriate technology and info systems.
- analyze, evaluate, and apply relevant information from a variety of sources.

Interpersonal

- show respect for the diverse opinions, values, belief systems, and contributions of others.
- interact with others in groups in ways that contribute to effective working relationships.

Personal

- manage the use of time and other resources to complete projects.
- take responsibility for one's own actions, decisions, and consequences.

Delivery Method

- Classroom: Course is delivered through scheduled synchronous teaching that may be face-to-face and/or virtual.
- Hybrid: Course combines scheduled synchronous and unscheduled asynchronous teaching.

Learning Activities

- Lectures
- Class Discussions
- Labs
- Group Work
- Research
- Self-Directed Learning
- Presentations
- Guest Speaker(s)
- Reflective Writing
- In-Class Exercises
- Case Studies

Evaluation Plan

Grading Scheme

A	80% - 100%
B	70% - 79%
C	60% - 69%
D	50% - 59%
F	0% - 49%

Evaluation Method	Value (%)
Assignments	40%
Applied Activities(G)	40%
Quizzes/Test/Exam	20%

ADDITIONAL INFORMATION

College

Academic Policies

Students must adhere to all academic policies. These are available on myCambrian as well as the College website at <https://cambriancollege.ca/about/official-documents-and-policies/academic-policies/>.

Prior Learning Assessment and Recognition (PLAR)

Students may contact the Pathways office at pathways@cambriancollege.ca

Transfer Credit

Students wishing to have courses from other programs or institutions assessed for equivalency and/or transfer credit should contact transfercredits@cambriancollege.ca.

Accessibility Services

Cambrian is committed to creating an inclusive learning environment where we support our students' success. Students who are registered with the Glen Crombie Centre can access accessibility and/or counselling services by:

1. Registering and scheduling an appointment through Clockwork in the Student Portal in myCambrian.
2. Contacting 705-566-8101 ext. 7420 or 7311, or
3. E-mailing accessibilityservices@CambrianCollege.ca

Testing Practices at Cambrian

Many courses include major tests and/or final exams. The practice at Cambrian requires that these types of test situations involve proctoring to ensure academic integrity. Online tests/exams may employ a proctoring service to enable you to take your exam from a location of your choosing within a period specified by your instructor. When you are taking an online test/exam, the proctoring service may capture your video, screen, audio and web surfing particulars to protect academic integrity. Cambrian College collects, uses, discloses and retains personal information in compliance with the Freedom of Information and Protection of Privacy Act (FIPPA). Your personal information is being collected under the authority of the Ontario Colleges of Applied Arts and Technology Act S.O. 2002, c.8, Sched. F. This information will be used for the purpose of administering a test/exam through an online proctoring service acting as an authorized agent of the College. Please refer to Cambrian's Confidentiality of Student Records Policy for more details. If you have any questions regarding the collection of your personal information, please contact Vice President Academic, Cambrian College, 1400 Barrydowne Rd., Sudbury ON P3A 3V8, 1-705-566-8101.

Wabnode Centre for Indigenous Services

Students are encouraged to come to Wabnode to meet our Elders and to access personal counselling or academic tutoring. Our team will also be able to provide students with information related to funding and scholarship and bursary opportunities for Indigenous students. We encourage students to come to Wabnode for cultural, sports and community activities. Students can also contact Wabnode at wabnode@CambrianCollege.ca

Copyright

Students may not submit work or partake in activities that infringe on the Canadian Copyright Act, or the related Cambrian College Fair Dealing Guidelines and Copying Guidelines. The guidelines may be viewed at: <http://CambrianCollege.libguides.com/copyrightinfo> or by contacting the Library for questions related to copyright: library@CambrianCollege.ca

Out-of-Class Assistance

The faculty will inform students of their Out-of-Class availability through the Learning Management System site (Moodle).

Audio/Visual Capture

Sounds and images from this class, and contributions made by a participant, virtually or in-person, are recorded under the authority of the Ontario Colleges of Applied Arts and Technology Act, 2002. The main purpose for these recordings is to allow students enrolled in this course, whether they attend the particular class in person, virtually, or at all, to review class content and activity.

These recordings may also be reviewed by faculty members to prepare future classes, to evaluate students, to collaborate in program/course development/review, or to provide feedback to students or other faculty. Any questions about this collection may be addressed to your respective Dean.

Equity, Diversity and Inclusivity

Cambrian is committed to building and preserving an equitable, diverse, and inclusive learning community where students, faculty, and staff may achieve their full potential in an environment characterized by equality of respect and opportunity. All students and employees have the right to live and work in an environment that is free from discrimination and harassment. Therefore, Cambrian College will not tolerate any form of discrimination or harassment in its employment, education, accommodation, or business dealings. For more information, please visit: <https://CambrianCollege.ca/about/official-documents-and-policies/equity-human-rights-and-accessibility/>

Tests and Evaluations

Students are required to write tests and/or complete evaluations as scheduled. Exceptions may be made in the event of an emergency or a sanctioned event (e.g., varsity sports, field trip, religious observances, etc.). In non-emergency situations, students are expected to contact the faculty member in the event that they can not be present for a test/evaluation to explain the reason for their absence. At the discretion of the faculty member, an alternate time for the test/evaluation may be allowed.

Attendance

Regular attendance is strongly encouraged as it contributes to student success. Some courses have specific lab/participation requirements. Students are expected to be aware of, and adhere, to these requirements.

Modifications to the Course Evaluation Plan

Under exceptional circumstances, Cambrian reserves the right to alter the Course Evaluation Plan. Upon approval from the Dean, faculty members will notify students of any modification to the Course Evaluation Plan and post to the course Learning Management System site (Moodle).

School of Business and Information Technology, Creative Arts, Design, Music and Hospitality Regulations**Lab Attendance**

Lab attendance is mandatory since lab grades include an evaluation of lab reports and lab performance. Lab performance can only be judged in person, requiring that students attend all scheduled labs.